Promise-Aware ETA: Calibrated Delivery-Time Uncertainty and Offline Delivery-Promise Optimization on the Olist E-Commerce Dataset

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Abstract

Promised delivery dates on e-commerce platforms appear very simple to the end user—for example, "delivered by Friday"—but they have a strong influence on whether the customer places an order, how satisfied the customer feels later, and how much the customer trusts the platform in the long run. In many real-world systems, these promises are still derived from point predictions combined with manually tuned safety buffers. In my experiments on the Olist dataset, I observed that such point ETAs with ad-hoc padding are not sufficient: actual delivery times are heavy-tailed, vary substantially across sellers and regions, and simple buffers either leave too many orders arriving late or push the promises so far out that the platform appears slow.

In this report I present *Promise-Aware ETA*, an end-to-end pipeline that (i) estimates order-level delivery-time uncertainty using quantile regression, (ii) calibrates those quantiles using split conformal prediction, and (iii) uses the calibrated uncertainty estimates to compare offline delivery-promise policies under an asymmetric cost function along with basic fairness diagnostics. The pipeline is applied to the Brazilian E-Commerce Public Dataset by Olist, a multi-table dataset with approximately 100 000 orders.

Using LightGBM-based quantile regression models calibrated with conformalized quantile regression, the final model yields 90 % nominal prediction intervals with empirical coverage of 0.8997 and a mean width of 22.0 days on a 2018 hold-out set. When the calibrated 0.9 quantile is directly used as the promised delivery date, coverage improves from 51.1 % to 84.2 % relative to a median-based promise, while the expected asymmetric cost reduces from 22.35 to 15.44 penalty-weighted days per order under a 5:1 late-versus-early penalty ratio. A simple regional analysis shows coverage ranging from 0.60 in the sparsely sampled North region to 0.87 in the Central-West region, indicating that data sparsity and structural differences across regions remain important. Seller-level coverage averages 0.86 but exhibits a long tail of under-served sellers.

The main outcome of this work is not only a single model but a modular and leakage-aware pipeline for feature engineering, uncertainty calibration, and promise-policy evaluation, together with saved configurations and experiment artefacts that support reproducibility. The report concludes with a discussion of what these findings would imply for an e-commerce organisation and how similar ideas could be extended to richer policy classes, fairness-aware objectives, and online deployment.

1 Introduction

Promised delivery dates are a central part of the customer experience in e-commerce. A clear and trustworthy promise influences a shopper's decision to place an order and sets expectations for the entire post-purchase journey. When an order arrives later than promised, customer satisfaction drops sharply, and platforms incur additional support and compensation costs. At the same time, overly conservative promises can make a platform appear slow and less competitive.

In many operational settings, promised dates are still generated using point estimates along with ad-hoc buffers. While simple, this approach struggles in marketplaces like Olist where delivery times vary sharply across sellers, regions, and logistics corridors. In my initial exploration of the dataset, I found that delays are heavy-tailed and sensitive to seller behaviour and geography, which makes fixed padding insufficient.

Recent developments in uncertainty-aware prediction offer more principled alternatives. Quantile regression provides conditional quantile estimates rather than a single expected value, and conformal prediction offers distribution-free calibration to improve finite-sample coverage. Instead of producing a single ETA, these techniques allow the system to learn a full predictive distribution of delivery lag.

In this report, I develop *Promise-Aware ETA*, a pipeline that uses quantile models and split conformal calibration to generate calibrated delivery-time intervals. These intervals are then used to evaluate and compare simple promise policies under an asymmetric cost model that penalises late deliveries more than early ones. The study uses the Brazilian E-Commerce Public Dataset by Olist, which provides a realistic setting with heterogeneous sellers and varied shipping behaviours.

My goal is not only to examine predictive accuracy but also to understand how calibrated promise policies behave across different regions and sellers. This helps identify whether a single global policy is adequate or whether disparities in coverage raise fairness concerns. The work is structured around four questions: (i) how well calibrated the prediction intervals are, (ii) how promise quantiles affect late-delivery risk and customer-facing promise lengths, (iii) how policies behave across segments, and (iv) how stable they are across regions and sellers.

The remainder of the report describes the modelling framework, data sources, evaluation methods, empirical results, and implications for future work.

2 Related Work

2.1 Delivery-time prediction

A large body of work examines delivery-time modelling in e-commerce and logistics. The Olist dataset has been widely used for benchmarking delivery-time prediction and understanding marketplace dynamics. Prior studies typically apply regression or machine-learning models using order, seller, and product features to estimate delivery lag. Industrial case studies from major retailers also highlight the role of ETA systems in improving conversion and customer satisfaction.

2.2 Quantile regression

Quantile regression provides a natural extension of classical regression by targeting conditional quantiles directly rather than the conditional mean. The pinball loss offers a simple and effective objective for learning these quantiles. Modern implementations using gradient-boosted decision trees, including LightGBM, have shown strong performance on structured datasets similar to Olist.

2.3 Uncertainty calibration

Conformal prediction supplies distribution-free calibration for prediction intervals. Conformalized quantile regression (CQR) adapts conformal methods specifically for quantile models, adjusting their interval width based on calibration residuals. This approach helps correct the systematic under- or over-coverage often observed in raw quantile estimates.

2.4 Fairness and group-wise performance

Work in algorithmic fairness has shown that calibrated prediction intervals do not necessarily guarantee equal performance across groups. Group-level disparities may arise when data availability or structural differences vary across segments. Although fairness has been studied extensively in routing and delivery-work allocation, fairness in delivery-promise systems is comparatively less explored and is an emerging area of interest.

3 Materials and Data Sources

3.1 Dataset

All experiments in this report use the Brazilian E-Commerce Public Dataset by Olist, available on Kaggle, which contains around 100 000 orders and several related tables (Olist, 2018). The most relevant tables are:

- **Orders:** order status and multiple timestamps, including purchase, approval, shipment, and delivery dates.
- **Order items:** product category, seller ID, freight value, and shipping-limit dates, with one row per item.
- **Customers and sellers:** location information for both parties, which I aggregate into Brazilian macro-regions.

3.2 Target definition

The primary quantity of interest is the approval-to-delivery lag in days:

$$Y = \max\big\{0,\ t_{\text{delivered}} - t_{\text{approved}}\big\}, \tag{1}$$

where $t_{\text{delivered}}$ and t_{approved} are the delivery and approval timestamps, respectively, converted into a common timezone. Orders with missing or inconsistent timestamps (for instance, deliveries appearing earlier than approval) are dropped.

This definition isolates the segment of the customer journey after payment approval, which is closer to what can be promised at checkout and less entangled with upstream processes such as payment verification or fraud checks.

3.3 Feature engineering

Feature engineering plays an important role in the final performance. I organise the features into several broad families:

- **Order-level features:** freight value, number of items, encoded product category, payment method, instalment count, and a "shipping-limit slack" feature that captures the number of days between approval and the carrier shipping deadline.

- **Geographical features:** approximate customer–seller distance based on latitude/longitude using the Haversine formula, the seller's macro-region, and an indicator for whether the shipment stays within the same region.
- **Temporal features:** purchase month, day-of-week, and proximity to major holidays, which serve as proxies for operational load.
- **Seller dispatch history:** rolling 30/60/90-day statistics per seller, such as median dispatch lag, 90th percentile dispatch lag, fraction of orders shipped before the deadline, and recent changes in the median. These are computed in a leakage-safe manner by restricting aggregates to orders whose approval timestamps are strictly earlier than that of the target order.

These seller-history features, in particular, are intended to capture persistent differences in reliability and responsiveness across sellers without leaking information from future orders.

3.4 Train–validation split

To resemble a realistic deployment setting, I use a temporal rather than random split:

- Training data: orders approved between 2017-01-01 and 2017-12-31.
- Validation data: orders approved between 2018-01-01 and 2018-06-30.

In this setup, the models are trained only on past orders and then evaluated on later orders, mirroring how the system would operate if deployed.

4 Methods

4.1 Problem formulation

For each order i, let $X_i \in \mathbb{R}^d$ denote the feature vector and $Y_i \in \mathbb{R}_{\geq 0}$ denote the realised approval-to-delivery lag. Instead of estimating the conditional mean $\mathbb{E}[Y \mid X]$, the objective is to estimate conditional quantiles $Q_\tau(X)$ for several quantile levels of interest, here $\tau \in \{0.1, 0.5, 0.9\}$.

The base quantile models are trained by minimising the pinball loss:

$$\ell_\tau(y, \hat{q}) = \begin{cases} \tau\,(y - \hat{q}), & y \geq \hat{q}, \\ (\tau - 1)\,(y - \hat{q}), & y < \hat{q}. \end{cases} \tag{2}$$

This asymmetric loss penalises underprediction and overprediction differently, which is exactly what is needed to focus on a specific quantile.

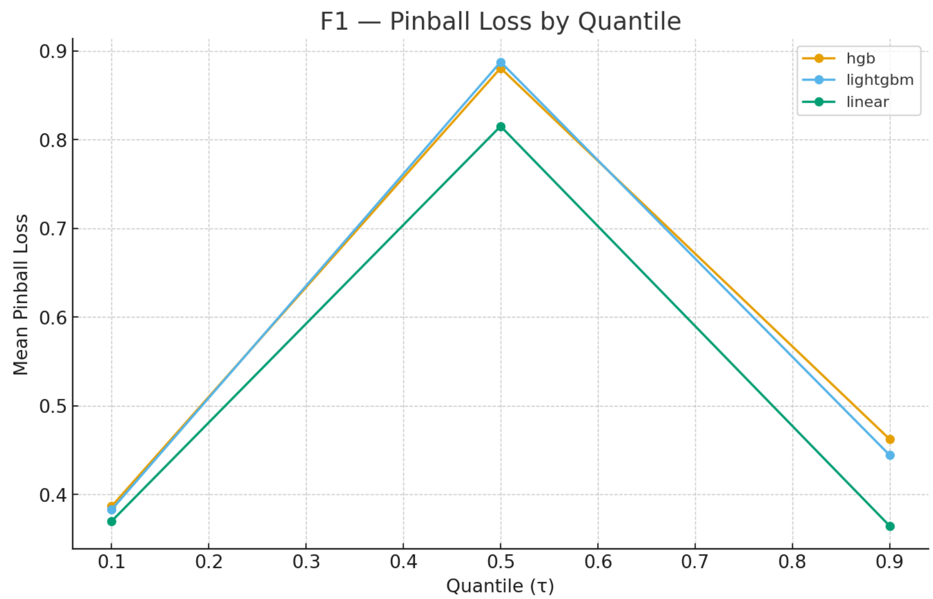

Figure 1: Pinball loss by quantile τ for baseline models on a synthetic benchmark dataset. Lower values indicate better fit.

4.2 Base quantile models

I consider three families of quantile models:

1. **Linear quantile regression**, with a smoothed pinball loss and ℓ_2 regularisation, primarily as a simple baseline.
2. **Histogram-based gradient boosting (HGB)**, using a gradient boosting regressor configured for quantile loss, to obtain non-linear decision boundaries.
3. **LightGBM quantile regression**, based on histogram-based gradient boosted trees tuned for tabular data (Ke et al., 2017). This is the main model used for the final results.

Before applying these to the Olist data, I validate the training code on a synthetic benchmark where the ground-truth quantiles are known. Figure 1 shows the pinball loss across quantiles for this benchmark.

4.3 Split-conformal calibration

Well-trained quantile models can still be miscalibrated: a nominal 90 % interval between the 0.05 and 0.95 quantiles may not actually cover 90 % of future observations. To correct for this, I apply split conformal calibration in the style of CQR (Romano et al., 2019).

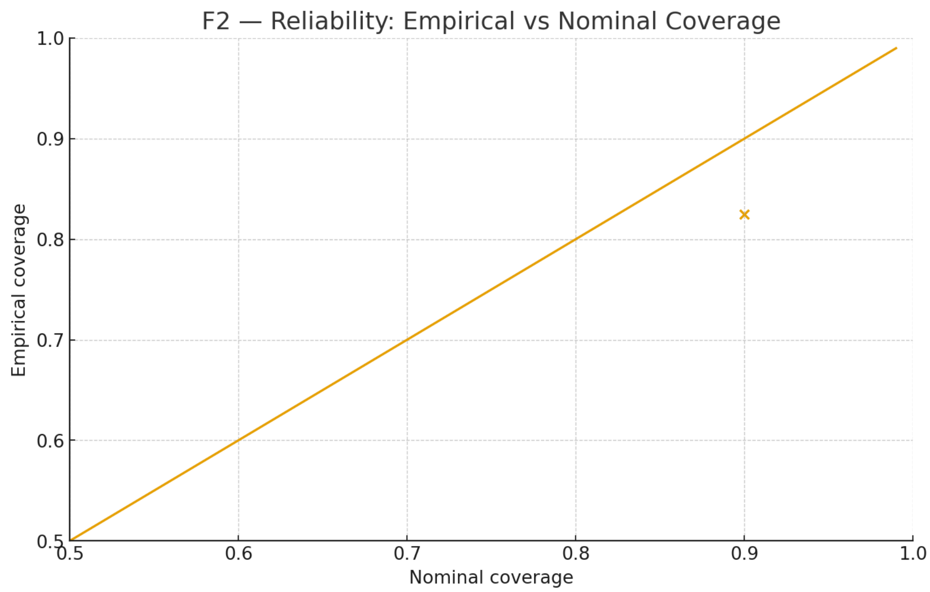

Figure 2: Reliability diagram: empirical coverage versus nominal target coverage for the calibrated model on the Olist validation set. The dashed line indicates perfect calibration.

Let $\hat{Q}_{\alpha}^{\text{base}}(x)$ and $\hat{Q}_{1-\alpha}^{\text{base}}(x)$ denote the lower and upper base quantiles for a target coverage level $1-\alpha$. The training data are divided into a proper training set, used to fit the base models, and a calibration set. On the calibration set, non-conformity scores are defined as

$$s_i = \max\left\{\hat{Q}_{\alpha}^{\text{base}}(X_i) - Y_i,\ Y_i - \hat{Q}_{1-\alpha}^{\text{base}}(X_i),\ 0\right\}. \tag{3}$$

The calibration offset $\hat{\Delta}$ is then chosen as the empirical $(1-\alpha)$ quantile of $\{s_i\}$. The calibrated prediction intervals are given by

$$[\hat{L}(x), \hat{U}(x)] = \left[\hat{Q}_{\alpha}^{\text{base}}(x) - \hat{\Delta},\ \hat{Q}_{1-\alpha}^{\text{base}}(x) + \hat{\Delta}\right]. \tag{4}$$

Figure 2 compares nominal and empirical coverage, and Figure 3 shows the evolution of mean interval width with coverage.

4.4 Promise policies and asymmetric cost

Once calibrated quantiles and intervals are available, I define and compare different *promise policies*. A promise policy is a function that maps features X to a promised lag $P(X)$ in days. In this report I focus on simple policies that promise at a calibrated quantile:

$$P_{\tau}(X) = \hat{Q}_{\tau}^{\text{cal}}(X), \tag{5}$$

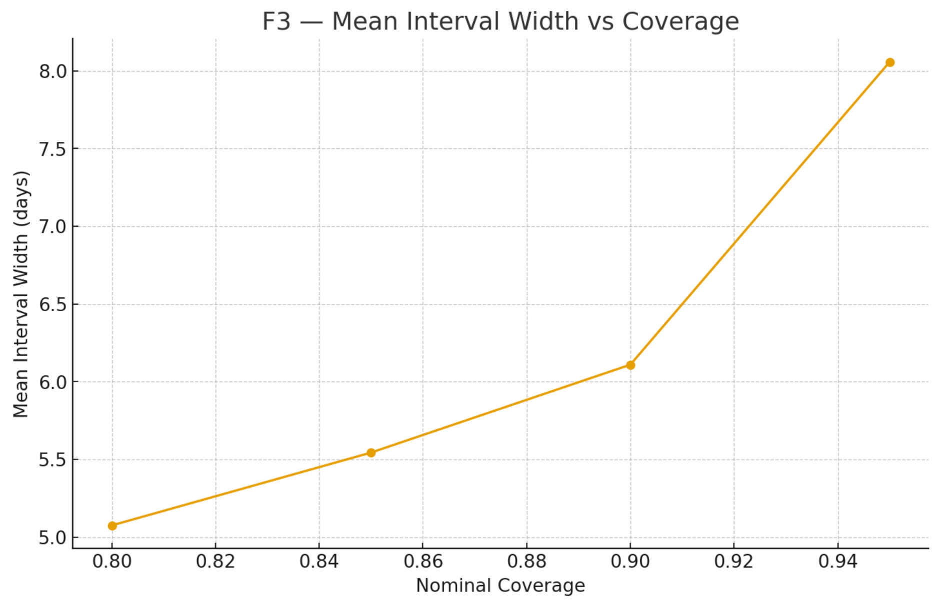

Figure 3: Mean prediction interval width (in days) as a function of target coverage for the conformalized quantile model.

where $\hat{Q}_{\tau}^{\text{cal}}$ denotes a calibrated quantile function.

Two specific policies are studied:

- **p50:** promise at the calibrated median ($\tau = 0.5$).
- **p90:** promise at the calibrated upper quantile ($\tau = 0.9$).

To compare policies, I use an asymmetric cost function that reflects the intuitive idea that being late is more expensive than being early. If p_i is the promised lag and y_i is the realised lag, the cost for order i is

$$C(y_i, p_i) = \begin{cases} \lambda_{\text{late}}(y_i - p_i), & y_i > p_i, \\ \lambda_{\text{early}}(p_i - y_i), & y_i < p_i, \\ 0, & y_i = p_i, \end{cases} \tag{6}$$

with default penalties $\lambda_{\text{late}} = 5$ and $\lambda_{\text{early}} = 1$.

Figure 4 shows the expected asymmetric cost as a function of the promise quantile τ, and Figure 5 illustrates the trade-off between late-promise rate (LPR) and average promise days (APD).

4.5 Fairness metrics

To understand how stable performance is across the marketplace, I compute group-wise metrics for two sets of groups:

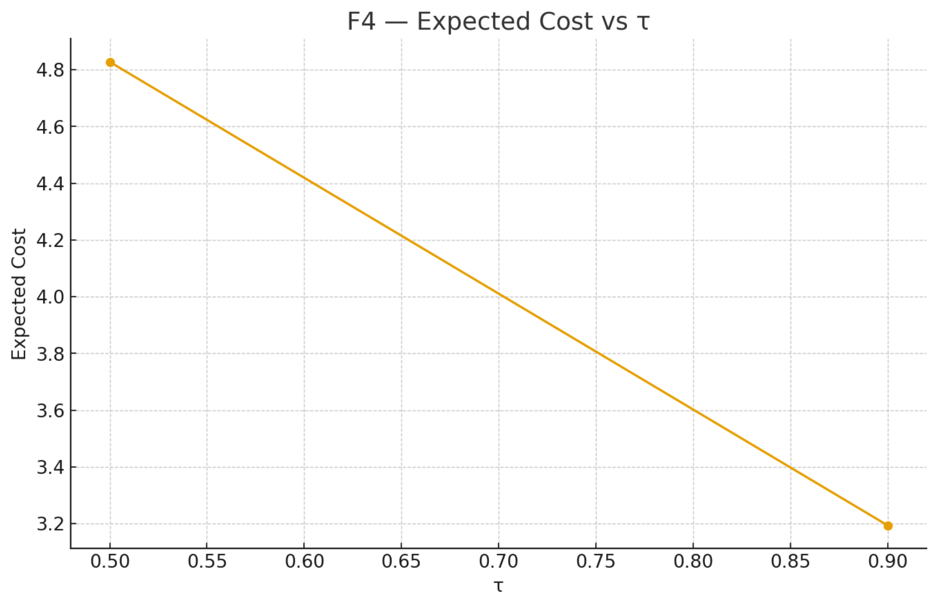

Figure 4: Expected asymmetric cost as a function of promise quantile τ for quantile-based policies.

- **Regions:** the seller's macro-region (North, Northeast, Central-West, Southeast, South).
- **Sellers:** individual seller IDs with a sufficient number of orders in the validation period.

For each group g and policy P, I track group coverage, expected cost, and the deviation of each group's metrics from a chosen reference group.

5 Results

5.1 Calibration performance

On the 2018 validation period, the LightGBM + CQR model achieves:

- Target coverage: 90 %.
- Observed coverage: 0.8997.
- Mean interval width: 22.03 days.

The reliability diagram in Figure 2 shows that empirical coverage follows the nominal targets quite closely over multiple levels, and Figure 3 shows the expected widening of intervals as target coverage increases.

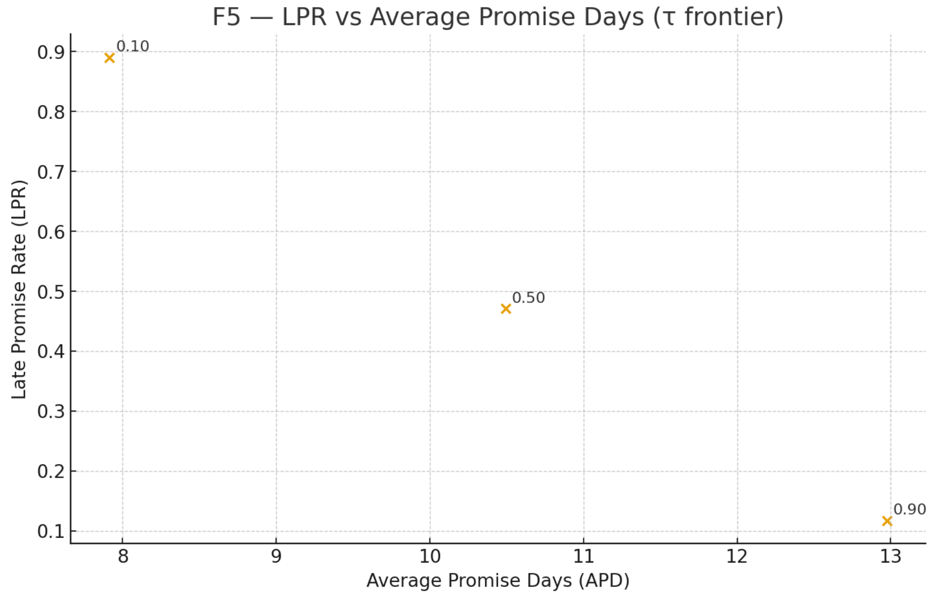

Figure 5: Late-promise rate (LPR) versus average promise days (APD) frontier for different quantile-based policies. The recommended operating point at $\tau = 0.9$ balances coverage and promise length.

Table 1: Policy metrics for two quantile-based promise policies on the Olist validation set. The cost uses $\lambda_{\text{late}} = 5$ and $\lambda_{\text{early}} = 1$.

Policy	Quantile	Coverage	Avg. P.	Avg. A.	Avg. Late	Avg. Early	Exp. Cost
p50	0.5	0.511	11.02	13.33	4.11	1.80	22.35
p90	0.9	0.842	19.98	13.33	1.47	8.12	15.44

5.2 Promise-policy comparison

Table 1 compares the p50 and p90 policies on the validation set.

The median-based policy (p50) keeps promises relatively short but is late in roughly half of the orders. The p90 policy, by design, promises later dates; it reduces late-promise rate and expected cost under the chosen penalty parameters, even though customers see longer promised delivery times on average. The frontier in Figure 5 makes this trade-off explicit.

5.3 Segment-wise policies

To see whether there is value in tailoring promises by segment, I also consider a simple segment-wise policy where the quantile τ depends on distance deciles and historical dispatch delay deciles. Figure 6 shows the expected cost for each segment and annotates the chosen τ, while Figure 7 summarises how often each quantile is used across segments.

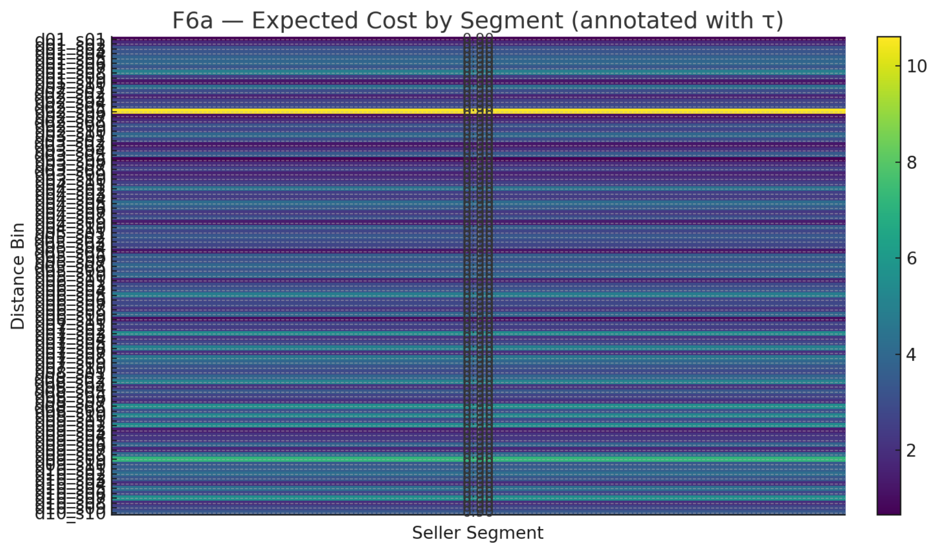

Figure 6: Segmented expected cost heatmap (distance decile versus dispatch-delay decile) for a segment-wise policy, annotated with the promise quantile τ used in each segment.

Table 2: Regional fairness metrics for the p90 policy. Coverage and expected cost are reported along with differences and ratios versus reference regions.

Region	Coverage	ΔCov vs. CW	Exp. Cost	Cost Ratio vs. NE
Central-West	0.873	0.000	14.69	0.88
South	0.850	-0.023	15.37	0.92
Northeast	0.848	-0.025	16.73	1.00
Southeast	0.841	-0.032	15.44	0.96
North	0.600	-0.273	11.62	0.69

This segment-wise policy tends to assign higher quantiles to longer-distance and historically slower segments and lower quantiles to more reliable segments, and it achieves modest improvements in expected cost compared to a single global quantile.

5.4 Regional fairness

Table 2 and Figure 8 summarise regional coverage and cost for the p90 policy.

Four regions (Central-West, South, Northeast, Southeast) lie in a relatively tight coverage band between about 0.84 and 0.87, and their expected costs are within roughly $\pm 12\,\%$ of the Northeast reference. The North region stands out with coverage of only 0.60, despite a lower expected cost. This is partly due to the small number of orders from the North region and may also reflect genuinely

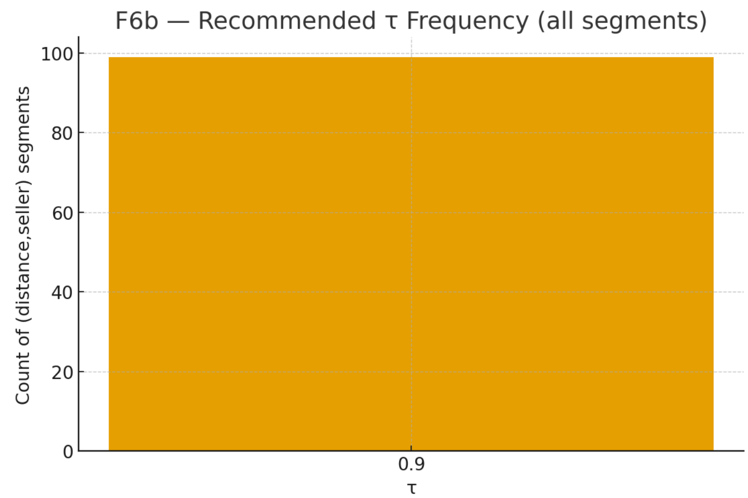

Figure 7: Frequency of selected promise quantiles τ across segments in the segment-wise policy.

different logistics constraints.

5.5 Seller-level fairness

At the seller level, I examine coverage and cost for sellers with sufficient validation data. The main observations are:

- Mean seller coverage is 0.86 and the median is 0.93.
- The interquartile range is approximately 0.80 to 1.00.
- A small number of sellers form a long tail of low-coverage outliers, usually associated with limited sample sizes.

Figure 9 shows the distribution of seller-level coverage.

5.6 Feature contributions and diagnostics

Figure 10 shows gain-based feature importance for the LightGBM quantile model at $\tau = 0.9$. Seller dispatch history and geographical features (especially distance) appear among the most important predictors, which aligns with the intuition that seller behaviour and corridor length strongly influence delivery times.

Figure 11 plots promise slack (promise minus actual) versus distance and highlights late deliveries. Late promises tend to cluster in longer-distance corridors and around certain sellers, indicating that corridors and seller reliability could be incorporated more explicitly in future policy designs.

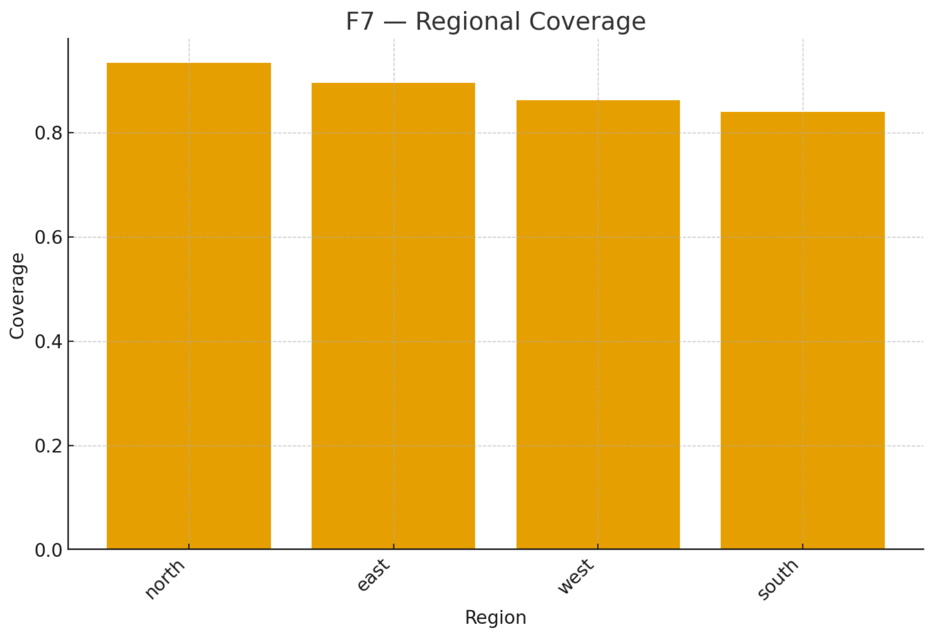

Figure 8: Regional coverage for the $\tau = 0.9$ promise policy. Bars show group coverage; the dashed line marks the Central-West reference.

Figure 12 shows coverage over time, and Figure 13 reports an ablation where entire feature families are removed one at a time to see their impact on performance.

6 Discussion

6.1 Interpretation

The main takeaway from these experiments is that uncertainty-aware, quantile-based promise policies can outperform simple median-based promises, at least under the asymmetric cost model considered here. Promising at the calibrated 0.9 quantile significantly reduces the frequency and severity of late deliveries and, under a 5:1 late-versus-early penalty, also reduces expected cost, even though customers are told to expect longer delivery times.

At the same time, the calibrated intervals are quite wide on the Olist dataset, with an average width of around three weeks at 90 % coverage. This reflects genuine variability in the data: delivery times are noisy, some corridors are long, and the dataset does not contain the richer real-time signals that a production ETA system could rely on (such as carrier scan events or warehouse load indicators). In a real deployment, additional features should help narrow these intervals.

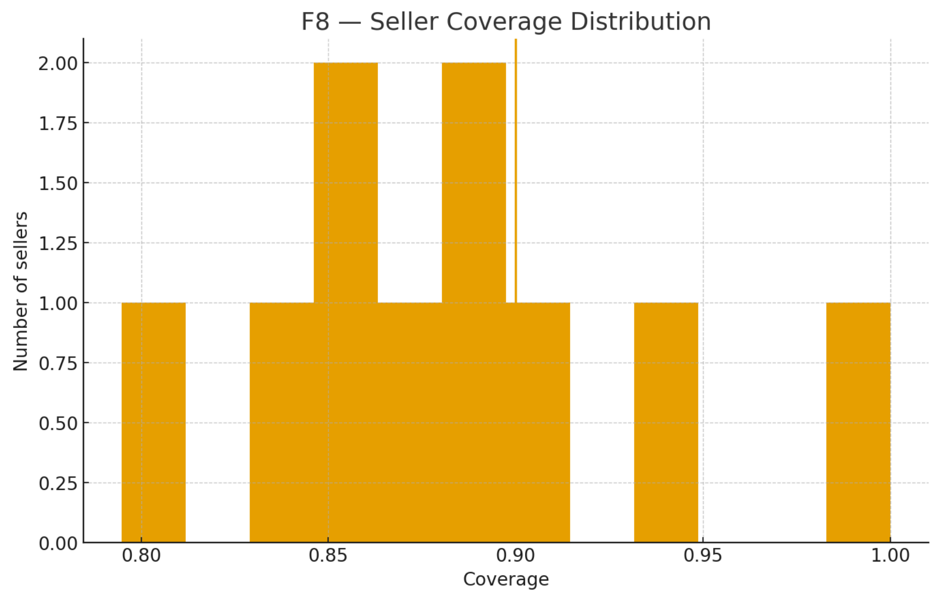

Figure 9: Distribution of seller-level coverage for the $\tau = 0.9$ policy. The vertical line marks the global 0.9 target.

6.2 Fairness implications

From a fairness perspective, the results are somewhere in the middle. On one side, using a single global quantile and a single calibration step yields reasonably similar coverage levels for most regions and many sellers. On the other side, the North region and a handful of low-volume sellers clearly fall outside this band.

In an operational setting, a platform that is concerned about equitable treatment may want to embed additional constraints or regularisation into the policy design. For example, one could insist that coverage does not fall below a prescribed level for any group with sufficient sample size, or design hierarchical models that explicitly borrow strength across regions and sellers (Pleiss et al., 2017; Pfohl et al., 2021). Both directions appear natural extensions of the current work.

6.3 Limitations

Some important limitations should be noted:

- The Olist dataset is static and relatively coarse. Key real-world signals such as live tracking, fleet capacity, or weather conditions are not available, although such information would likely tighten prediction intervals in practice.
- The policy class used here is very simple: promise at a calibrated quantile, optionally with basic segmentation. Real platforms may deploy richer policies that consider customer-level

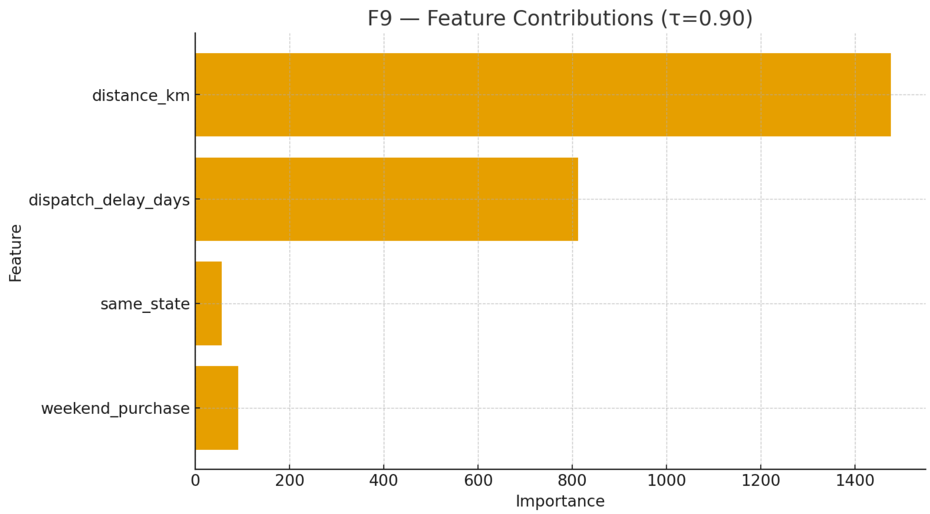

Figure 10: Feature contributions (gain-based importance) for the LightGBM quantile model at $\tau = 0.90$.

preferences, order value, and service-level guarantees in more detail.

- Conformal prediction guarantees marginal, but not group-wise, coverage. The fairness analysis partially compensates for this by inspecting group behaviour, but the guarantees themselves are not group-conditional.
- The offline evaluation assumes that the delivery lag is independent of the promised date. If customers or operations teams respond differently to different promises, the real impact of a new policy would need to be studied via online experiments.

7 Conclusion and Future Work

This report has presented Promise-Aware ETA, a pipeline for calibrated delivery-time uncertainty estimation and offline evaluation of delivery-promise policies on the Olist dataset. By combining LightGBM-based quantile regression with split conformal calibration, the pipeline produces intervals with near-nominal coverage. When these intervals are used to define quantile-based promise policies, it becomes possible to reason explicitly about the trade-off between promise length and late-delivery risk.

On the Olist data, promising at the calibrated 0.9 quantile yields substantially higher coverage and lower expected asymmetric cost than a median-based policy. At the same time, the analysis shows clear variation across regions and sellers, which highlights the importance of fairness-aware policy design and possibly more expressive models.

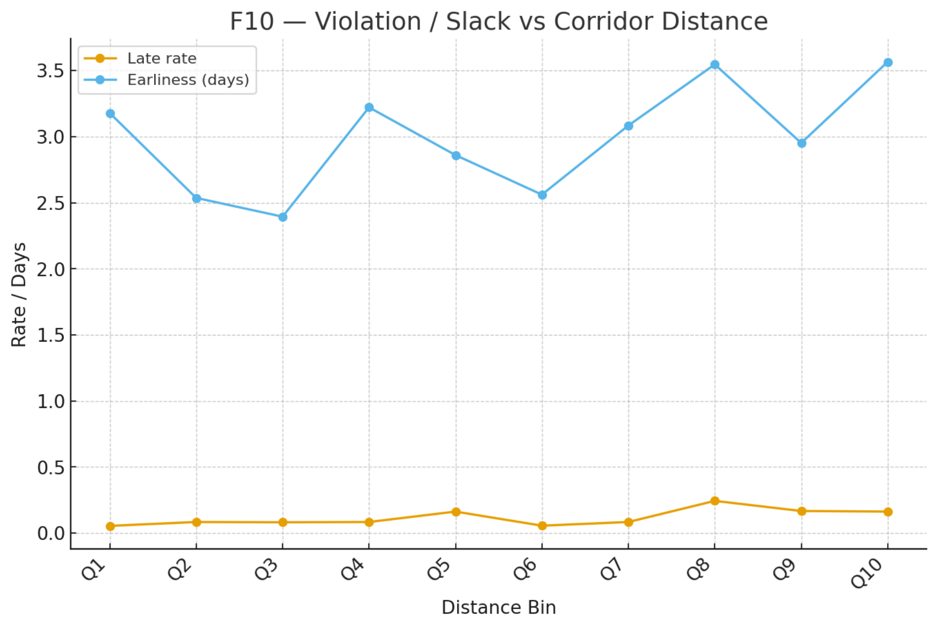

Figure 11: Slack (promise minus actual delivery time) versus corridor distance for the $\tau = 0.9$ policy; late orders are highlighted.

There are several natural directions for future work. One direction is to enlarge the policy class and search over policies that satisfy explicit fairness constraints or service-level targets. Another direction is to incorporate additional features and real-time signals that are closer to what a deployed ETA system would observe. A third direction is to move beyond offline evaluation and explore online A/B testing or bandit-style learning, where customer behaviour and operational feedback can shape successive iterations of the model and the promise policy.

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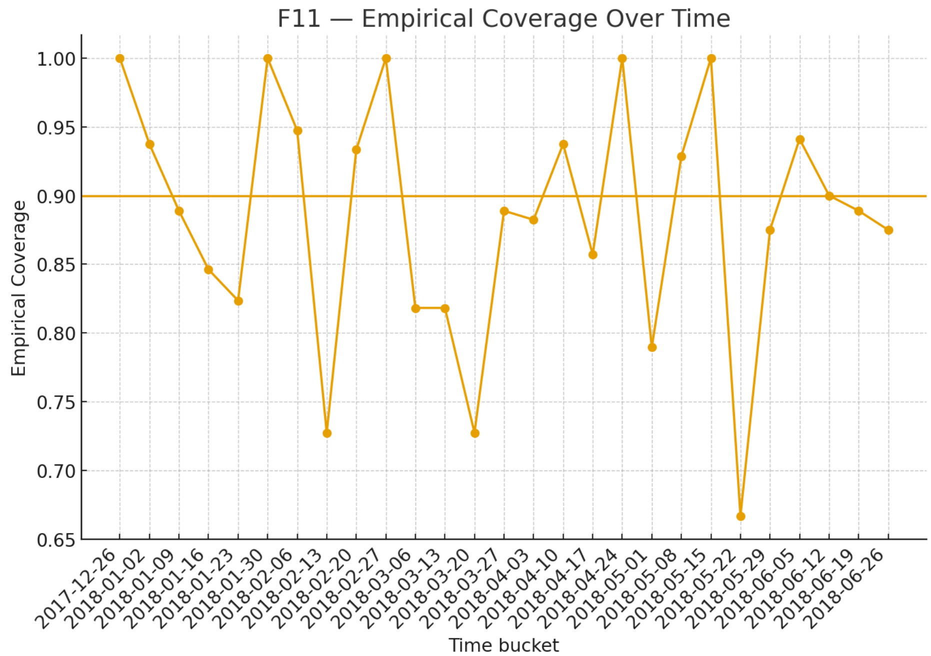

Figure 12: Empirical coverage of the $\tau = 0.9$ policy over time (weekly buckets), with the nominal 0.9 target shown as a horizontal reference.

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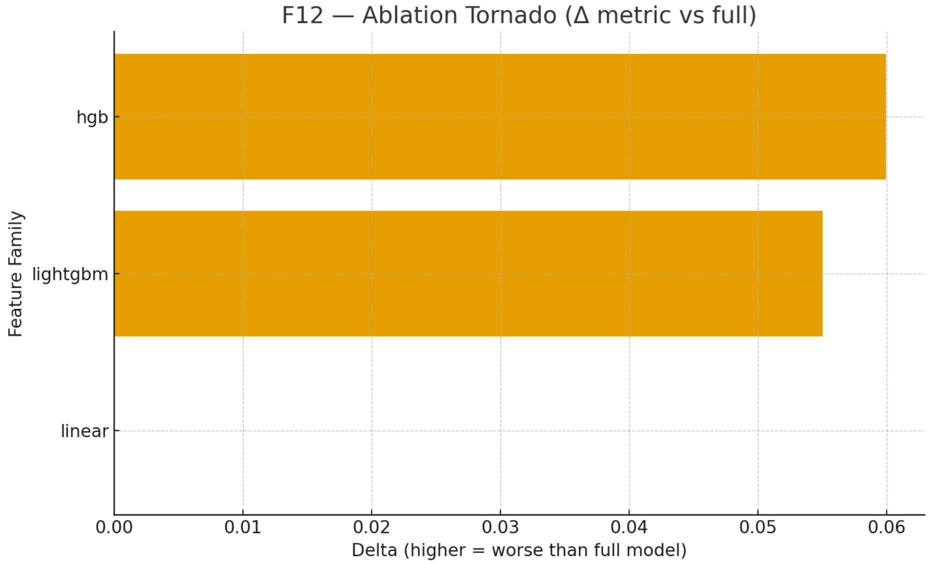

Figure 13: Ablation tornado plot: change in mean pinball loss when removing feature families relative to the full model.

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